ASSIGNMENT No: 1

TITLE: Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Problem Statement

Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

Understand the theoretical concepts, implement them in Java programs, analyze outputs, and apply the concepts to solve practical programming problems.

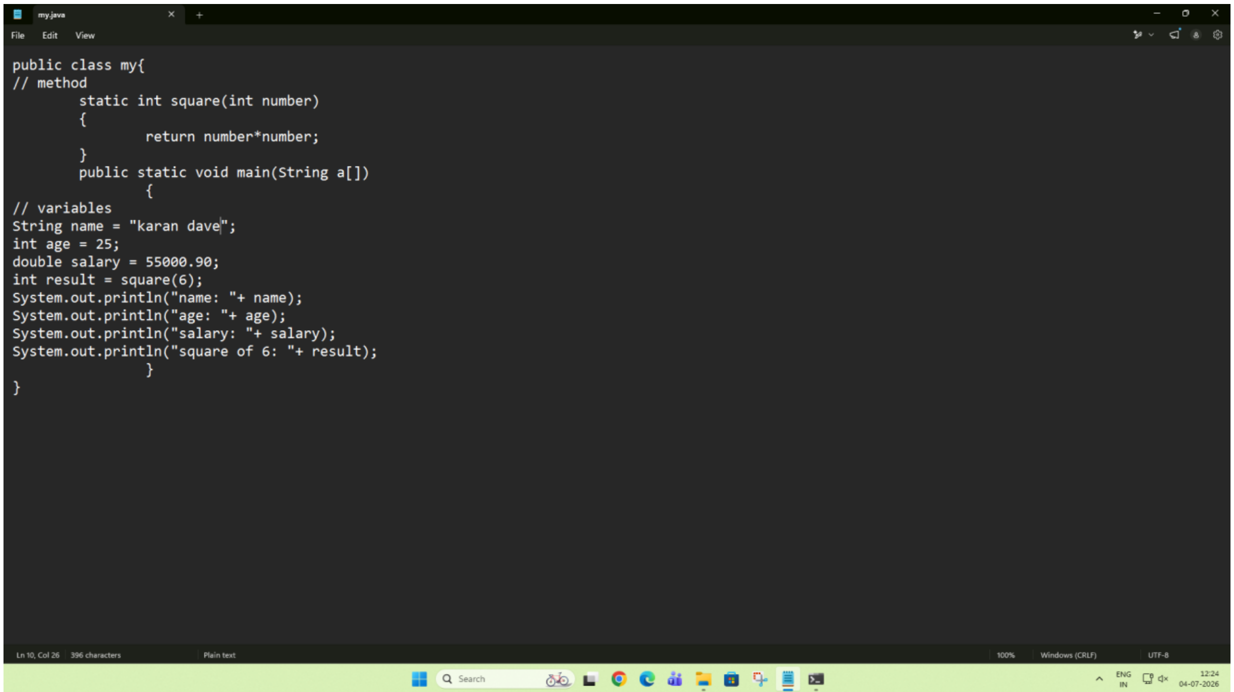
Theory

Java environment setup, JDK/JRE/JVM, variables, methods and constructors. forms an important component of Java programming. Students should understand the underlying concepts, architecture, advantages, use cases, and implementation methodology. The exercise should focus on understanding the design principles, coding practices, execution flow, and practical applications of these concepts in software development. Detailed study should include syntax, APIs, execution process, and real-world applications.

Algorithm

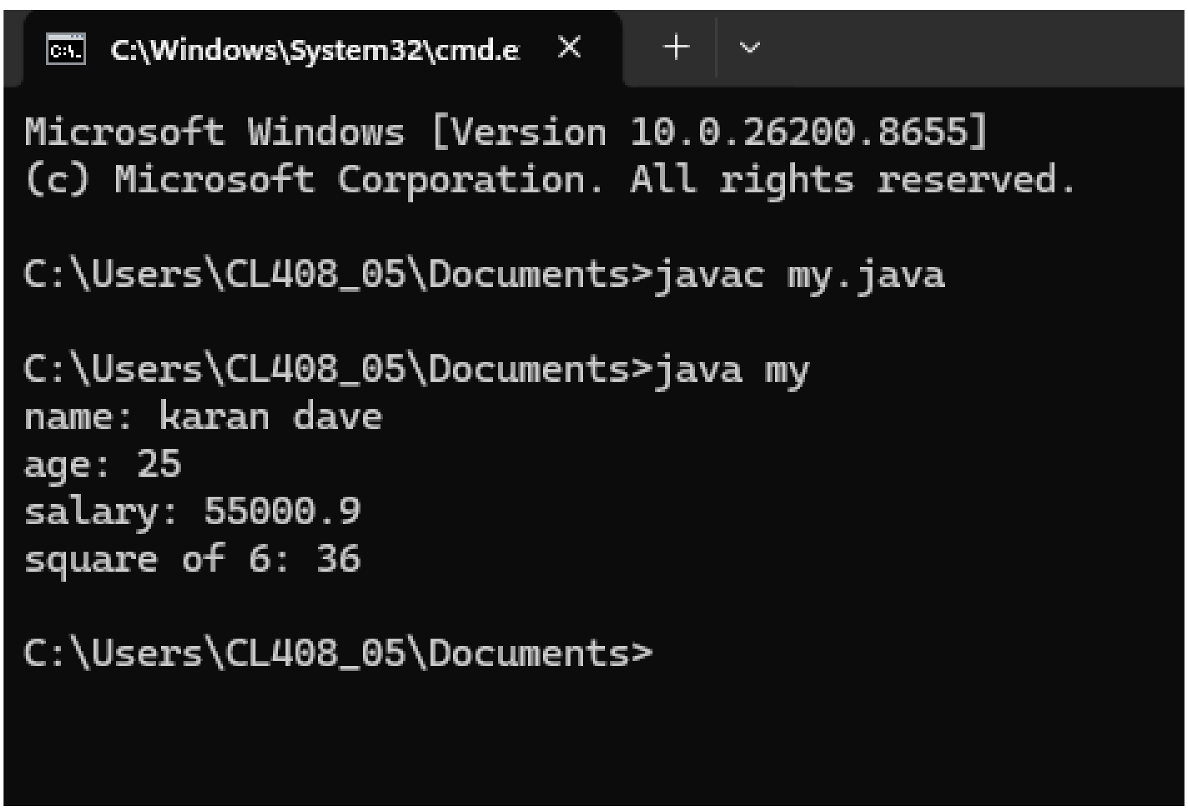
1. START.
2. Create a class named "my".
3. Define a static method "square (int number)".
4. Calculate square by multiplying "number * number".
5. Return square.
6. Define main () method.
7. Declare & initialize variables-
   a) String name = "Karan Dave".
   b) int age = 25
   c) double salary = 55000.90
8. Call square(6) method and store in variable "result".
9. Display name, age, salary, result.
10. STOP.

Code




```java
public class my{
// method
	static int square(int number)
	{
		return number*number;
	}
	public static void main(String a[])
		{
// variables
String name = "karan dave";
int age = 25;
double salary = 55000.90;
int result = square(6);
System.out.println("name: "+ name);
System.out.println("age: "+ age);
System.out.println("salary: "+ salary);
System.out.println("square of 6: "+ result);
		}
}
```

Output




```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\CL408_05\Documents>javac my.java

C:\Users\CL408_05\Documents>java my
name: karan dave
age: 25
salary: 55000.9
square of 6: 36

C:\Users\CL408_05\Documents>
```

Exercise

1. Explain the main concept used in Java environment setup, variables and methods.

   It is the process of preparing a computer to develop & run programs.
   JDK provides tools to develop & compile programs.
   JVM executes Java bytecode.
   Variables store datatypes like int, char, strings.
   Methods are blocks of code that perform tasks.

2. What are the advantages of using Java variables and methods?

   a) It is the process of preparing a computer to develop & run programs.
   b) JDK provides tools to develop & compile programs.
   c) JVM executes Java bytecode.
   d) Variables store datatypes like int, char, strings.
   e) Methods are blocks of code that perform tasks.

3. How can this implementation be improved?

   a) Use meaningful names of variables and methods.
   b) Add comments.
   c) Avoid duplicate code by method.
   d) Use exception handling to manage runtime errors.
   e) Apply OOP concepts.

4. What are the real-world applications?

   a) Banking systems - Accounts, deposits, withdrawals.
   b) Student management systems - Student records.
   c) Apps - Handle user data & functions.
   d) E-commerce - Process orders, payments, customer care.
   e) Healthcare - Patient records, appointments.
   f) Enterprise - Business operations, databases.